APPLIED STATISTICS, PROBABILITY THEORY AND MATHEMATICAL STATISTICS

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Basic notions of statistics

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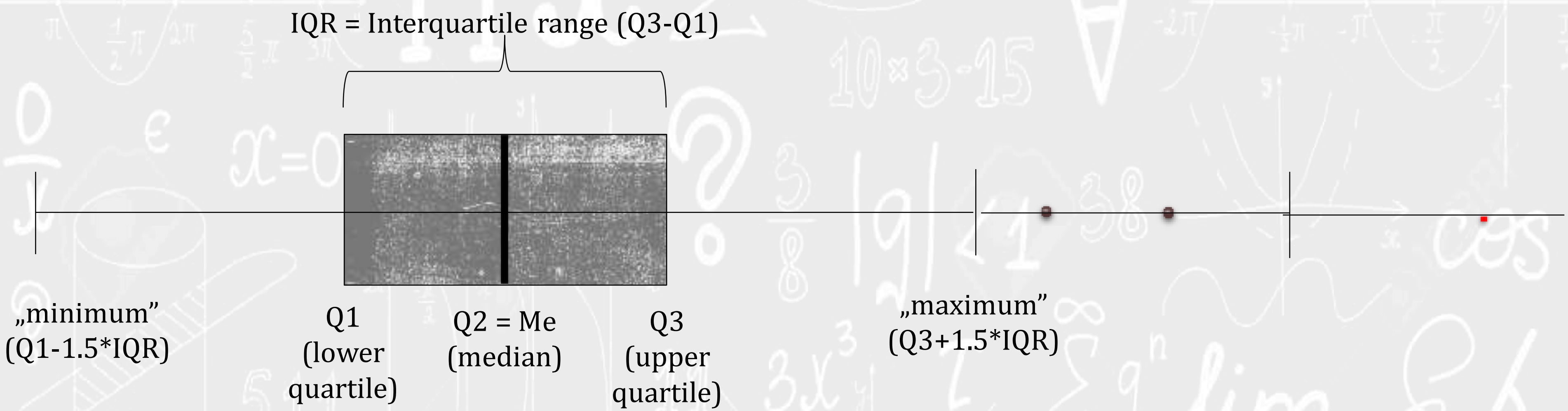
Boxplot



Outlier: $Y_i < Q1 \ - 1.5 \cdot IQR$ or $Y_i > Q3 + 1.5 \cdot IQR$

Extreme values : $Y_i < Q1 \ - 3 \cdot IQR$ or $Y_i > Q3 + 3 \cdot IQR$

Exercise 16.1 (M1)

Drink it tea is sold in cartons of tea bags. The weight, in grams, of the content of each carton in any batch is supposed to be 200. A quality controller had bought five cartons and weighted their contents. The results, in grams, are the following:

196, 202, 198, 197, 190.

Find the mean and the variance of the sample and plot the empirical cumulative distribution function.

Exercise 16.1 (M1)

```
data=[196, 202, 198, 197, 190];
mean(data)
var(data)
```

Exercise 16.1 (M1)

```
data=[196, 202, 198, 197, 190];
data=sort(data);
min_x=2*min(data)-mean(data); max_x=2*max(data)-mean(data);
axis([min_x max_x -0.1 1.1])
hold on
plot(min_x:0.1:min(data), zeros(length(min_x:0.1:min(data))),'b')
for i=1:(length(data)-1)
    plot((data(i):0.1:data(i+1)),
i/length(data)*ones(length(data(i):0.1:data(i+1))),'b')
end
plot(max(data):0.1:max_x, ones(length(max_x:-0.1:max(data))),'b')
```

Exercise 16.1 (M1) (Save this code in a script, named ecdf2)

```
function eloszfv = ecdf2(data)
data=sort(data);
min_x=2*min(data)-mean(data);
max_x=2*max(data)-mean(data);
axis([min_x max_x -0.1 1.1])
hold on
plot(min_x:0.1:min(data),zeros(length(min_x:0.1:min(data))),'b')
for i=1:(length(data)-1)    plot((data(i):0.1:data(i+1)),
i/length(data)*ones(length(data(i):0.1:data(i+1))),'b')
end
plot(max(data):0.1:max_x, ones(length(max_x:-0.1:max(data))),'b')
end
```

Exercise 16.1 (M1) (Save this code in a script, named ecdf21)

```
function elfv = ecdf21(data)
udata=unique(data);
data1=udata;
for i=1:length(udata)
data1(2,i)=sum(data==udata(i));
end
data=data1;
min_x=2*min(data(1,:))-mean(data(1,:));
max_x=2*max(data(1,:))-mean(data(1,:));
axis([min_x max_x -0.1 1.1])
hold on
plot(min_x:0.01:min(data(1,:)),
zeros(length(min_x:0.01:min(data(1,:)))),'b','LineWidth',2)
for i=1:(size(data,2)-1)

plot((data(1,i):0.01:data(1,(i+1))),sum(data(2,1:i)/sum(data(2,:)))*ones(lengt
h(data(1,i):0.01:data(1,(i+1)))),'b','LineWidth',2)
end
plot(max(data(1,:)):0.01:max_x, ones(length(max_x:-
0.01:max(data(1,:)))),'b','LineWidth',2)
end
```

Exercise 16.2 (M2)

The following 8 numbers were generated by a random number generator generating uniform random numbers on the (0; 1) interval.

0.18, 0.57, 0.82, 0.55, 0.63, 0.12, 0.91, 0.31.

Find the mean and the variance of the sample and plot both the empirical and the true cumulative distribution function.

Exercise 16.2 (M2)

```
data = [0.18, 0.57, 0.82, 0.55, 0.63, 0.12, 0.91, 0.31];
 mean(data)
 std(data)
 ecdf21(data)
 hold on
 x = -0.1:0.01:1.1;
plot(x,unifcdf(x),'r')
```

Cars

```
load carsmall

boxplot(MPG)
xlabel('All Vehicles')
ylabel('Miles per Gallon (MPG)')
title('Miles per Gallon for All Vehicles')

boxplot(MPG,Origin)
title('Miles per Gallon by Vehicle Origin')
xlabel('Country of Origin')
ylabel('Miles per Gallon (MPG)')

tbl = tabulate(Origin);
t = cell2table(tbl,'VariableNames', ...
   {'Value','Count','Percent'});
t.Value = categorical(t.Value)

bar(t.Value,t.Count)
xlabel('Country of Origin')
ylabel('Number of Cars')
```

Covid

```
covid = readtable('covid.csv');

head(covid)

covid.countriesAndTerritories = categorical(covid.countriesAndTerritories);

tabulate(covid.countriesAndTerritories)

hun = covid.cases(covid.countriesAndTerritories=="Hungary");

plot(flip(hun))
title('Number of COVID-19 cases in Hungary per day')

aust = covid.cases(covid.countriesAndTerritories=="Austria");

hold on
plot(flip(aust),'r')
title('Number of COVID-19 cases in Hungary and Austria per day')

n = min(length(aust),length(hun));
corr(hun(1:n),aust(1:n))
```

Literature

- Ágnes Baran: Mathematics for engineers 1. (Laboratory slides)
- Sándor Baran: Probability theory and statistics
- Fazekas István: Valószínűségszámítás, Debreceni Egyetemi Kiadó, 2009
- Fazekas István: Bevezetés a matematikai statisztikába, Debreceni Egyetemi Kiadó, 2009
- Matlab examples: https://www.mathworks.com/help/examples.html